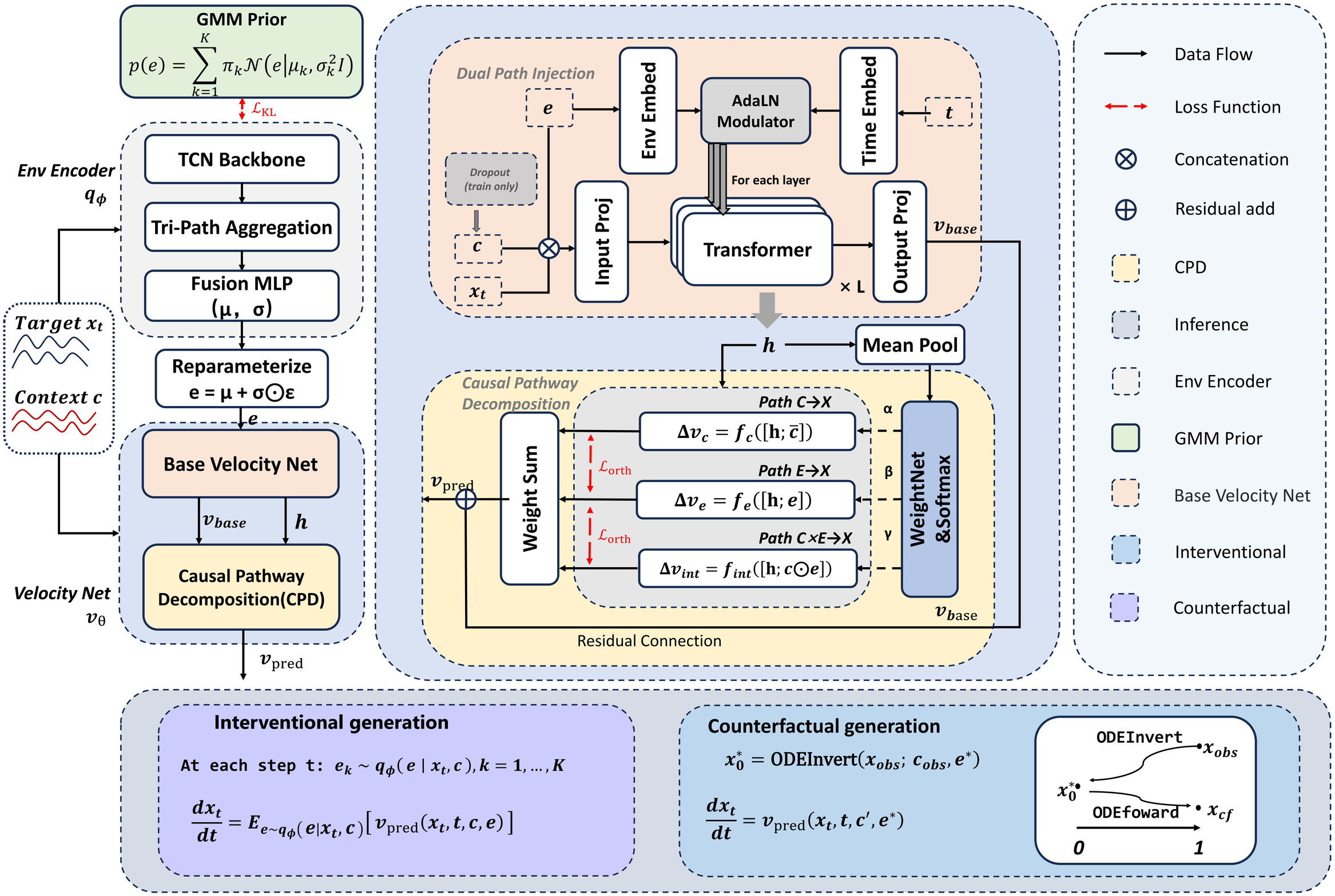
GMM Prior
$p(e) = \sum_{k=1}^{K} \pi_k \mathcal{N}(e|\mu_k, \sigma_k^2 I)$
$\mathcal{L}_{KL}$
Env Encoder
q_ϕ
TCN Backbone
Tri-Path Aggregation
Fusion MLP
(μ, σ)
Target x_t
Context c
Reparameterize
e = μ + σ⊙ε
e
Base Velocity Net
v_{base}
h
Causal Pathway Decomposition(CPD)
Velocity Net
v_θ
v_{pred}
Dual Path Injection
e
Dropout (train only)
c
x_t
Input Proj
Env Embed
AdaLN Modulator
Time Embed
t
For each layer
Transformer
× L
Output Proj
v_{base}
h
Mean Pool
Causal Pathway Decomposition
Path C→X
$\Delta v_c = f_c([\mathbf{h}; \bar{c}])$
Path E→X
$\Delta v_e = f_e([\mathbf{h}; e])$
Path C×E→X
$\Delta v_{int} = f_{int}([\mathbf{h}; c \odot e])$
$\mathcal{L}_{orth}$
$\mathcal{L}_{orth}$
α
β
γ
WeightNet &Softmax
Weight Sum
v_{pred}
v_{base}
Residual Connection
Data Flow
Loss Function
Concatenation
Residual add
CPD
Inference
Env Encoder
GMM Prior
Base Velocity Net
Interventional
Counterfactual
Interventional generation
At each step t: $e_k \sim q_\phi(e \mid x_t, c), k = 1, \ldots, K$
$\frac{dx_t}{dt} = E_{e \sim q_\phi(e|x_t, c)}[v_{pred}(x_t, t, c, e)]$
Counterfactual generation
$x_0^* = \text{ODEInvert}(x_{obs}; c_{obs}, e^*)$
$\frac{dx_t}{dt} = v_{pred}(x_t, t, c', e^*)$
ODEInvert
x_{obs}
x_0^*
ODEfoward
x_{cf}
0
1